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SECTION: **CSE C-1**

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Lab No. 9 – DataBases – Part1

Q1. Assume a table “Institutes” with institute_id, name, and no_of_courses are the fields. Create a web page that retrieves all the data from “Institutes” table displays only Institute names in the list box.

Forms.py

```
from django import forms
from .models import Institute

class InstituteSelectionForm(forms.Form):
    """Form to display institutes as a selectable list"""
    institute = forms.ModelChoiceField(
        queryset=Institute.objects.all(),
        widget=forms.Select(attrs={
            'class': 'form-control',
            'id': 'institute-select'
        })),
        label='Select Institute',
        empty_label='-- Choose an Institute --'
    )
```

Urls.py

```
from django.urls import path
from . import views
```

```
urlpatterns = [
    path("", views.institute\_list, name='institute_list'),
]
```

Urls.py

```
from django.contrib import admin
from django.urls import path, include
```

```
urlpatterns = [
    path('admin/', admin.site.urls),
    path("", include('institutes.urls')),
]
```

Settings.py

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'institutes',
]
```

```
MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]
```

```
ROOT_URLCONF = 'institute_project.urls'
```

```
TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [],
        'APP_DIRS': True,
        'OPTIONS': {
```

```

'context_processors': [
'django.template.context_processors.request',
'django.contrib.auth.context_processors.auth',
'django.contrib.messages.context_processors.messages',
],
},
},
]

WSGI_APPLICATION = 'institute_project.wsgi.application'

# Database
# https://docs.djangoproject.com/en/5.2/ref/settings/#databases

DATABASES = {
'default': {
'ENGINE': 'django.db.backends.sqlite3',
'NAME': BASE_DIR / 'db.sqlite3',
}
}

```

Models.py

```

from django.db import models

class Institute(models.Model):
    """Institutes table model"""
    institute_id = models.AutoField(primary_key=True)
    name = models.CharField(max_length=200)
    no_of_courses = models.IntegerField()
    created_at = models.DateTimeField(auto_now_add=True)

    class Meta:
        ordering = ['name']
        verbose_name_plural = "Institutes"

    def __str__(self):
        return f'{self.name} {self.no_of_courses} courses}'

```

Institute.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Institute Selector</title>
<style>
*{
margin: 0;
padding: 0;
box-sizing: border-box;
}

body {
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
background-color: #f5f5f5;
color: #333;
line-height: 1.6;
min-height: 100vh;
display: flex;
flex-direction: column;
}

.form-group {
margin-bottom: 1.5rem;
}

label {
display: block;
font-weight: 600;
margin-bottom: 0.5rem;
color: #2c3e50;
}

button {
width: 100%;
}
}
</style>
</head>
<body>
```

```

<header>
<div class="container">
<h1>📖 Institute Directory</h1>
</div>
</header>

<main>
<div class="container">
<!-- Info Box -->
<div class="info-box">
<p><strong>Welcome!</strong> Select an institute from the list below to view its details.</p>
</div>

<!-- Statistics -->
<div class="stats-grid">
<div class="stat-card">
<div class="stat-value">{{ total_institutes }}</div>
<div class="stat-label">Total Institutes</div>
</div>
</div>

<!-- Form Section -->
<div class="form-section">
<h2>Select an Institute</h2>
<form method="post" class="form">
{% csrf_token %}
<div class="form-group">
<label for="institute-select">{{ form.institute.label }}</label>
{{ form.institute }}
</div>
<div class="form-actions">
<button type="submit" class="btn-primary">View Details</button>
<a href="{% url 'institute_list' %}" style="padding: 0.75rem 1.5rem; background-color: #95a5a6; color:
white; text-decoration: none; border-radius: 4px; display: inline-block; cursor: pointer;">Clear
Selection</a>
</div>
</form>
</div>

<!-- Results Section -->
{% if selected_institute %}
<div class="result-section">
<h3>✓ Institute Details</h3>

```

```

<div class="result-details">
<div class="detail-item">
<div class="detail-label">Institute ID</div>
<div class="detail-value">#{{ selected_institute.institute_id }}</div>
</div>
<div class="detail-item">
<div class="detail-label">Institute Name</div>
<div class="detail-value">{{ selected_institute.name }}</div>
</div>
<div class="detail-item">
<div class="detail-label">Number of Courses</div>
<div class="detail-value">{{ selected_institute.no_of_courses }}</div>
</div>
</div>
</div>
{% endif %}

<!-- All Institutes List -->
<div class="institutes-list">
<h3>All Available Institutes</h3>
{% if institutes %}
{% for institute in institutes %}
<div class="institute-item">
<span class="institute-name">{{ institute.name }}</span>
<span class="institute-courses">{{ institute.no_of_courses }} courses</span>
</div>
{% endfor %}
{% else %}
<p style="color: #999; text-align: center; padding: 2rem;">No institutes available.</p>
{% endif %}
</div>
</div>
</main>

<footer>
<p>&copy; 2026 Institute Management System. All rights reserved.</p>
</footer>
</body>
</html>

```

Wsgi.py

```
import os

from django.core.wsgi import get_wsgi_application

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'institute_project.settings')

application = get_wsgi_application()
```

OUTPUT -

The screenshot displays the Django administration interface for the 'Institutes' model. The interface is in a dark theme. The sidebar on the left contains navigation links for 'AUTHENTICATION AND AUTHORIZATION' (Groups, Users) and 'INSTITUTES' (Institutes). The main content area shows a table of institutes with the following data:

INSTITUTE ID	NAME	NO OF COURSES	CREATED AT
9	BITS Pilani	92	March 25, 2026, 9:58 a.m.
4	Bangalore Institute of Science	65	March 25, 2026, 9:58 a.m.
6	Chennai Engineering College	88	March 25, 2026, 9:58 a.m.
2	Delhi University	120	March 25, 2026, 9:58 a.m.
8	ISM Dhanbad	45	March 25, 2026, 9:58 a.m.
1	Indian Institute of Technology	85	March 25, 2026, 9:58 a.m.
7	Kolkata Institute of Technology	72	March 25, 2026, 9:58 a.m.
10	Manipal Academy of Higher Education	110	March 25, 2026, 9:58 a.m.
3	Mumbai University	95	March 25, 2026, 9:58 a.m.
5	Pune Technical Institute	78	March 25, 2026, 9:58 a.m.

The table shows 10 institutes. A filter sidebar on the right allows filtering by 'No of courses' and 'By created at'.

Activity | Microsoft TeamsWPL_Lab9.docxWPL_Lab9.pdfInstitute SelectorSelect institute to change | DjanInstitute Directory

Welcome! Select an institute from the list below to view its details.

10
Total Institutes

Select an Institute

Select Institute

-- Choose an Institute --

View DetailsClear Selection

All Available Institutes

BITS Pilani	92 courses
Bangalore Institute of Science	65 courses
Chennai Engineering College	88 courses
Delhi University	120 courses
ISM Dhanbad	45 courses
Indian Institute of Technology	85 courses
Kolkata Institute of Technology	72 courses
Manipal Academy of Higher Education	110 courses
Mumbai University	95 courses
Pune Technical Institute	78 courses

Activity | Microsoft TeamsWPL_Lab9.docxInstitute SelectorSelect institute to change | DjanInstitute SelectorInstitute Directory

Welcome! Select an institute from the list below to view its details.

10
Total Institutes

Select an Institute

Select Institute

Manipal Academy of Higher Education (110 courses)

View DetailsClear Selection

✓ Institute Details

Institute ID	Institute Name	Number of Courses
#10	Manipal Academy of Higher Education	110

All Available Institutes

BITS Pilani	92 courses
Bangalore Institute of Science	65 courses
Chennai Engineering College	88 courses
Delhi University	120 courses
ISM Dhanbad	45 courses
Indian Institute of Technology	85 courses